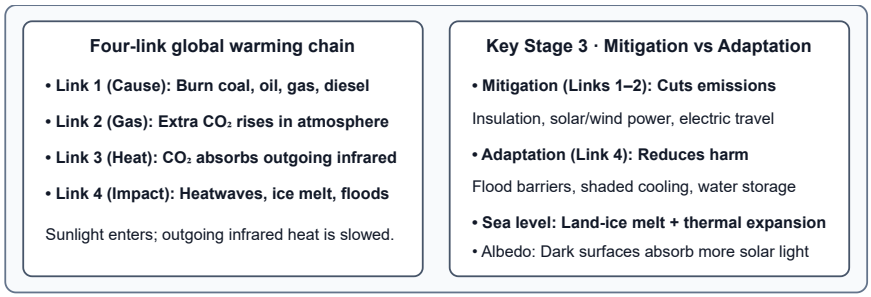
Reference sheets

Reference A · Four-link global warming chain and climate choices

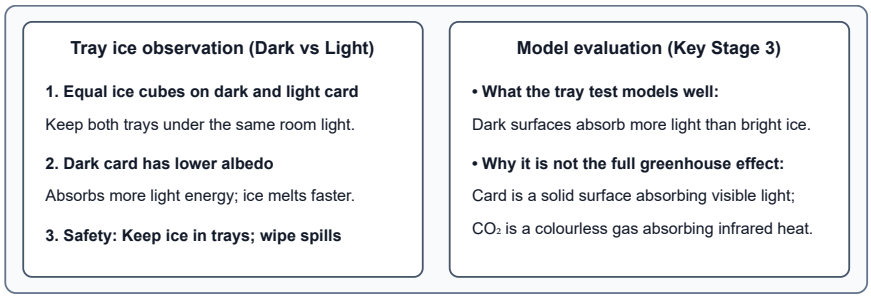


Four-link warming chain drawing. It shows fossil fuel burning and CO_2 infrared heat. It shows mitigation and adaptation.

Reference only. Do not write on this sheet. Follow Links 1 to 4 from cause to impact. Check which link each choice targets.

Chain link or choice.	What happens.	Key Stage 3 science mechanism.	Examples and limit.
Link 1 · Cause.	Power plants, heaters, and cars burn fossil fuels like coal, oil, and gas.	Burning fuel joins carbon with oxygen to make carbon dioxide (CO_2) gas.	Main human cause of extra greenhouse gas release.
Links 2 & 3 · Air & Heat.	Air CO_2 rises. More outgoing heat stays in the air.	Sun rays pass in. CO_2 and methane trap outgoing infrared heat.	The enhanced greenhouse effect slows heat leaving for space.
Link 4 · Climate impacts.	Global mean heat rises over 30 years or more.	Warmer air and seas bring heatwaves, floods, land-ice melt, and warm sea swelling.	Melting bright ice drops albedo so dark ground takes in more light.
Mitigation vs Adaptation.	Mitigation cuts gas at Links 1–2. Adaptation cuts harm at Link 4.	Mitigation: loft wraps, solar or wind power. Adaptation: flood gates, shade, water butts.	Limit: CO_2 stays in air for decades. Flood gates do not cut CO_2.

Reference B · Tray ice test (Dark card vs Light card) and model limits



Tray ice test drawing. It compares ice cubes on dark card and light card. It shows safety rules and limits.

Test feature.	**What we see on the tray.**	**Key Stage 3 reason and model limit.**
Equal ice on dark vs light card.	The ice on dark card melts faster under the same room light.	Dark card has lower albedo. It takes in more light and warms faster.
Why it is a partial model only.	It shows how dark ground takes in more sun rays when ice melts.	Limit: Dark card taking in light is not clear CO_2 gas trapping infrared heat.
Safety in the room.	Keep ice in the trays on the desk. Use room light, not a heater.	Melted ice can make a wet slip risk. Wipe any spill right away.

Made by Matt

Tier pack · Aspire

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Use the arrival drawing. Tell incoming sun rays apart from outgoing Earth heat in the greenhouse effect.

Help. Compare the sun ray arrow with the outgoing infrared heat arrows.

2. Use the four-link chain in the arrival drawing. How does the extra human greenhouse effect differ from the natural one?

Help. Link burning fossil fuel at Link 1 to extra CO_2 at Link 2.

3. Why does sea level rise at Link 4 when global mean heat rises?

Help. Join land-ice melt with warm sea water swelling.

4. Why can adaptation at Link 4 (like shaded cool zones) never replace mitigation at Link 1?

Help. Think about Links 2 and 3 if fuel burning never stops.

You do tasks

1. Use Reference A and Reference B to explain the enhanced greenhouse gas effect. Link fossil fuel burning, carbon dioxide, infrared heat, albedo, and warm sea rise.

Staff I ☐ S ☐ C ☐

Help. Join Links 1 to 4 on Reference A with albedo on Reference B.

2. Why must a town join mitigation (electric buses) with adaptation (flood gates and shade)? State one limit of each choice. **Staff I ☐ S ☐ C ☐**

Help. Compare Link 1 mitigation with Link 4 adaptation on Reference A.

3. A pupil claims: 'Ice melted faster on dark card in our tray test. That proves carbon dioxide gas is black.' Use Reference A and Reference B to fix this error. **Staff I ☐ S ☐ C ☐**

Help. Tell dark card albedo apart from clear CO_2 gas trapping infrared heat.

Optional participation-loop prompt. Share your four-link chain or choice note with an adult. Hear one feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. Explain the enhanced greenhouse effect. Distinguish incoming solar radiation from outgoing infrared radiation.

2. Why does a flood barrier or shade (Link 4 adaptation) not stop global warming?

3. How does melting ice create an albedo loop alongside thermal sea water expansion?

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Aspire

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt